

Library Management System

Installation & Setup Manual

Welcome to the installation guide for the Library Management System. This application is divided into three main components:

- **The Database** (MySQL)
- **The API Server (Back-end)** (Node.js & Express)
- **The Client Applications (Front-ends):**
 - User Portal (React/Vite)
 - Admin Dashboard (React/Vite)

🔧 Prerequisites

Before starting, ensure you have the following software installed on your machine:

- **Node.js** (version 16 or higher recommended): Download Node.js
- **MySQL Server:** Download MySQL (or use a local environment like XAMPP, WAMP, MAMP)

Step 1: Database Setup

1. Open your MySQL manager (e.g., phpMyAdmin, MySQL Workbench, or command line).
2. Create a new database for the application. For example: `library_db`.
3. Navigate to the database folder of your project (`My app/database`).
4. **Import the schema:** execute the `dblibrary.sql` script in your new database to create the required tables.

5. **(Optional) Import the sample data:** execute the `seed.sql` script to populate the database with initial test data (books, users, etc.).

Step 2: Back-end Setup (API Server)

The API server handles requests between the database and the front-end applications.

1. Open a terminal (or command prompt) and navigate to the back-end folder:

```
cd "path/to/My app/back-end"
```

2. Install the project dependencies:

```
npm install
```

3. Inside the back-end folder, create a file named `.env` to configure the database connection. Add the following content (modify according to your local setup):

```
DB_HOST=localhost  
DB_USER=root  
DB_PASSWORD=your_mysql_password  
DB_NAME=library_db
```

4. Start the development server:

```
npm start
```

Note: You should see a message saying *"Our Server is listening !!!"*. The API will be accessible at `http://localhost:3000`.

Step 3: User Portal Setup (Front-end)

This application is the public interface for library members.

1. Open a new terminal and navigate to the user portal folder:

```
cd "path/to/My app/front-end/login page/login-page"
```

2. Install the dependencies:

```
npm install
```

3. Start the web application:

```
npm run dev
```

The user application will be accessible in your browser, typically at `http://localhost:5173`.

Step 4: Admin Dashboard Setup (Front-end)

This application is exclusively for library staff to manage daily operations.

1. Open a new terminal and navigate to the admin dashboard folder:

```
cd "path/to/lib"
```

2. Install the dependencies:

```
npm install
```

3. Start the web application:

```
npm run dev
```

The admin dashboard will be accessible in your browser (Vite will display the local URL, for example, `http://localhost:5174`).